

A Lean 4 tribute to Cédric Villani:

a discrete Brunn–Minkowski inequality, fully formalized,
and a stability criterion that turned out not to be formalizable yet

Xavier Callens

SocrateAI-Scientific-QuantumFluids

<https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>

September 2026

Abstract

We report a small Lean 4 formalization offered as a tribute to Cédric Villani, built from a literature survey of his freely available 2025–2026 work and lecture notes rather than from the one result of his that already has a machine-checked proof (nonlinear Landau damping, formalized independently by Bedrossian). The survey found that Mathlib already proves the Kullback–Leibler Data Processing Inequality for a Markov kernel, so we state its three-line specialization to an invariant measure – the discrete, relative-entropy form of Boltzmann’s H-theorem – explicitly *not* as a contribution. The headline result is a full, **sorry-free** proof of Theorem 1 of Ollivier and Villani’s *curved Brunn–Minkowski inequality on the discrete hypercube* (SIAM J. Discrete Math. 2012) at zero curvature: for nonempty A, B in the Hamming cube with midpoint set M , $\#A \cdot \#B \leq (\#M)^2$, by the crossover-coding injection $A \times B \hookrightarrow M \times M$ the paper describes. We also report a companion literature gate, run before any code, on a separate proposal from this library’s kinetic-theory side: to identify Pomeranchuk’s thermodynamic stability criterion for a Landau Fermi liquid with a Penrose-type dynamical one for the two-dimensional Landau kinetic equation. The literature establishes that correspondence for an analogous three-dimensional system, but not, so far as we found, for the two-dimensional case; and the numerical parameters needed even to state the two-dimensional version are not published anywhere we could reach. We stopped rather than derive either from scratch, and report why.

1 Introduction

This library already leans on three large machine-checked developments: OpenAI’s Navier–Stokes/Euler formalization [1], Anthropic’s formalization of Fermat’s Last Theorem, evaluated as a source of general analysis [2], and Bedrossian’s Lean 4 formalization of nonlinear Landau damping [3], which subsumes one small lemma we had proved independently [4]. Asked to build a foundation formalizing Cédric Villani’s own work as a tribute, the obvious first move was *not* to duplicate that last result, and the second was to find out what, among his output, is both self-contained and not already done.

What is and is not claimed. *New:* a full proof of Ollivier–Villani’s Theorem 1 at zero curvature (§4). *Not a contribution, stated as such:* a corollary of Mathlib’s own Data Processing Inequality (§3). *Not attempted, named:* the curved case of the same theorem, and everything listed in §5. *Stopped at a literature gate, before any code:* a proposed link between the Pomeranchuk and Penrose stability criteria (§6).

2 What is already there

We read thirteen freely available items: Villani’s two solo papers of 2025–2026, *Fisher Information in Kinetic Theory* [5] and *On the kinetic Fokker–Planck equation in curved geometry* [6]; Imbert–Silvestre–Villani [7]; Mouhot–Villani on Landau damping [8] and its short notes; *Hypocoercivity* [9]; Desvillettes–Mouhot–Villani on Cercignani’s conjecture [10]; Lott–Villani on Ricci curvature via optimal transport [11, 12]; Ollivier–Villani on a discrete Brunn–Minkowski inequality [13]; *Optimal Transport, Old and New*; and *H-theorem and beyond*.

Against this, and against Mathlib as pinned by this project (not a different checkout, and not from memory), we found: `InformationTheory.klDiv` together with the Kullback–Leibler Data Processing Inequality for a measurable map, a sub- σ -algebra, and a Markov kernel (`klDiv_comp_right_le`), Gibbs’ inequality (`mul_log_le_klDiv`), and the chain rule; the Lévy–Prokhorov metric and Prokhorov’s theorem; Gaussian measures and Brownian motion. We did *not* find Wasserstein distance, Kantorovich–Rubinstein duality, a log-Sobolev or Talagrand inequality (the name occurs only in a bibliography entry), Bakry–Émery theory, Otto calculus, the Boltzmann equation, Fisher information, a Poincaré inequality, a heat kernel, the Fokker–Planck equation, or Ricci curvature via optimal transport. Outside Mathlib, J. K. Miller’s Lean 4 development of mean-field Vlasov theory (arXiv:2607.08986) proves W_1 , couplings, the gluing lemma and Kantorovich–Rubinstein duality for Polish spaces from Mathlib alone – the natural next dependency for anyone formalizing optimal transport in this ecosystem, though we do not use it here.

3 A corollary, explicitly not a contribution

For a Markov kernel κ and a measure ν invariant under it, $\kappa \circ_m \nu = \nu$, the Data Processing Inequality applied to μ and ν gives $\text{klDiv}(\kappa \circ_m \mu, \nu) = \text{klDiv}(\kappa \circ_m \mu, \kappa \circ_m \nu) \leq \text{klDiv}(\mu, \nu)$: relative entropy to an invariant measure is nonincreasing along a Markov chain – Boltzmann’s H-theorem in the form Cover and Thomas give it [14], and the discrete skeleton of the entropy methods Villani surveys in *H-theorem and beyond*. The Lean proof is three lines, `klDiv_nonincreasing_to_invariant` in `Villani.lean`; the mathematical content is entirely Mathlib’s own `klDiv_comp_right_le`. We include it because it is the right first step in this direction and because it connects this file to `ZeroSound.lean`’s Fermi-liquid kinetic theory inside one project, not because it is ours.

4 Ollivier–Villani’s Theorem 1, at zero curvature

Ollivier and Villani compare two notions of discrete Ricci curvature on the hypercube $X = \{0, 1\}^N$ with the Hamming metric [13]. Along the way they prove a discrete analogue of the classical Brunn–Minkowski inequality: for nonempty $A, B \subseteq X$, with M the set of *midpoints* of A and B (a point m with $d(m, a) + d(m, b) = d(a, b)$ and $d(m, a)$ within 1 of $d(a, b)/2$, for some $a \in A, b \in B$),

$$\ln \#M \geq \frac{1}{2} \ln \#A + \frac{1}{2} \ln \#B + \frac{K}{8} d(A, B)^2, \quad K = \frac{1}{2N}. \quad (1)$$

We formalize the $K = 0$ case, $\#A \cdot \#B \leq (\#M)^2$, which the paper itself isolates as §3 (“Brunn–Minkowski inequality without curvature”) before adding the curvature term. Its own proof is an explicit injection $A \times B \hookrightarrow M \times M$, which we reproduce faithfully – read from the paper, not guessed, and re-derived by hand before a line of Lean was written.

The injection. For $a, b \in X$ let $D(a, b) = \{i : a_i \neq b_i\}$, $r = \#D(a, b)$. For $c \subseteq D(a, b)$ define the *crossover* $\varphi_c(a, b)_i = a_i$ if $i \in c$, else b_i ; this is a midpoint of a, b exactly when $\#c$ is within 1 of $r/2$. For a *canonical* choice – $c_r(a, b)$, the $\lfloor r/2 \rfloor$ smallest indices of $D(a, b)$ under the order on $\text{Fin } N$, a function of $D(a, b)$ alone – the pair $\Phi(a, b) = (\varphi_{c_r}(a, b), \varphi_{D(a, b) \setminus c_r}(a, b))$ is recoverable:

one checks directly that $D(\varphi_c(a, b), \varphi_{D(a, b) \setminus c}(a, b)) = D(a, b)$ for *any* $c \subseteq D(a, b)$, so r is read off the two output midpoints as their own Hamming distance, which pins down c_r again, and then each coordinate of a, b is read off the two midpoints by whether it lies in c_r , $D(a, b) \setminus c_r$, or neither. This makes Φ injective, hence $\#(A \times B) \leq \#(M \times M)$.

<code>crossover, diff_crossover_left/right</code>	the crossover map and its two distances
<code>isMidpoint_crossover</code>	any size-appropriate crossover is a genuine midpoint
<code>takeHalf, card_takeHalf</code>	the canonical half, via Mathlib’s <code>Finset.orderEmbOfFin</code>
<code>encode, encodefst/snd_isMidpoint</code>	Φ , and that both components are midpoints
<code>diff_crossover_crossover_compl</code>	$D(\Phi(a, b)) = D(a, b)$, for any crossover split
<code>encode_injective</code>	Φ is injective (the case split above)
<code>card_sq_ge</code>	$\#A \cdot \#B \leq (\#M)^2$

Table 1: The load-bearing steps from the definitions to Theorem 1 at $K = 0$. The module’s axiom audit covers 16 theorems: 15 for this chain (including unfolding and distance lemmas not shown) and the entropy corollary of §3. Standard axiom footprint `{propext, Classical.choice, Quot.sound}`; accepted by both the Lean kernel and the independent `nanoda` kernel [15]; three negative controls (a stricter bound, a relaxed midpoint inequality, a strict Data Processing Inequality) all fail to compile.

Beyond first principles the proof uses only a set-difference cardinality identity, Mathlib’s canonical order embedding of a finite subset, and the cardinality bound for an injection between finite sets: nothing in it needed machinery this project did not already have on hand for other reasons.

5 Not attempted, and why

- **The curved case of the same theorem**, (1) with $K = 1/(2N)$. The paper’s own proof needs concentration of measure in the symmetric group S_r (its Lemma 4, via a quotienting argument onto the space of “crossovers”, its Proposition 5) – machinery we did not build. Left as a named open target rather than silently dropped.
- **The Fisher-information monotonicity results of [5, 7]**: Fisher information is nonincreasing along the spatially homogeneous Boltzmann equation. Tensorisation ($I(f \otimes f) = 2I(f)$) and the Cramér–Rao inequality look tractable (Mathlib has the needed integration-by-parts machinery on $\mathbb{R}^d$); the full monotonicity theorem is a genuine analytic argument, not attempted.
- **A discrete H-theorem for a finite-velocity collision kernel**, by the classical four-fold symmetrisation of the entropy production, was our first plan, abandoned mid-proof when §3’s three-line route through existing Mathlib content turned out to give a sharper and more general fact directly.
- **Everything genuinely out of reach today**: the continuous collisional Boltzmann H-theorem (no entropy production functional in Mathlib), hypocoercivity in Villani’s abstract Hilbert-space form [9], log-Sobolev and Talagrand inequalities (no Wasserstein distance, no Gaussian concentration machinery), the Lott–Villani–Sturm curvature-dimension condition [11, 12], and nonlinear Landau damping, already done [3].

6 A literature gate that stopped a proposal before it was code

This library’s kinetic-theory side had independently proved, in Lean, that undamped zero sound in a Fermi liquid with a single Landau parameter F_0^s exists if and only if $F_0^s > 0$, in both two and three dimensions [4]. A natural next step, chosen to bring the Villani side and the Godfrin side

of this project together, was to formalize a stronger, dynamical statement: that the *linearized* Landau kinetic equation is stable – no root of its dispersion relation at $\text{Im } \omega > 0$ – if and only if $F_0^s > -1$, the range where Pomeranchuk’s thermodynamic stability criterion holds, and to enumerate the corresponding roots with certified interval arithmetic for real ^3He at the pressures Godfrin et al. measured, in both bulk and the two-dimensional film their neutron-scattering experiment used [16].

Following this project’s rule to search before writing code, we ran that search first.

Stopped. The thermodynamic Pomeranchuk criterion $F_0^s > -1$ is standard and well sourced [17]. Its equivalence to the dynamical, Penrose-type statement is not an established theorem for the case at hand. The closest match, Kolomeitsev and Voskresensky [18], obtains exactly that correspondence by explicit dispersion analysis – but for a three-dimensional scalar-interaction nuclear-matter model, not named after Penrose, and not the two-dimensional case this project targets. Formalizing the statement as we had proposed it would need an *original* two-dimensional dynamical derivation, which is outside what this project formalizes. Separately, the numerical parameters the proposal needed even to run – areal-density-dependent Landau parameters for the two-dimensional ^3He film Godfrin et al. actually measured – are not available to us: their Nature paper is paywalled, and no companion paper surfaced a usable table within the search budget. Both halves of the proposal failed their own pre-registered kill criteria, and neither was implemented.

We report this because a stopped proposal, with the reason recorded, is data about the method, not an absence of one: three prior topological proposals in this project were likewise stopped or refuted after being pre-registered [19], and the discipline that catches them is the same one that let §4 be checked with confidence once it was written.

7 Conclusion

A literature survey aimed at “formalize something of Villani’s” found more value in what was already proved elsewhere (Bedrossian) or already in the library (Mathlib’s Data Processing Inequality) than in inventing a target; the one genuinely new piece of formal content here is a co-authored theorem old enough, and self-contained enough, that its proof could be read from the paper and checked, rather than guessed at. The stopped proposal of §6 is offered as the same kind of evidence this project’s other negative results are: not a discovery, but a fact about where the discipline’s cost is paid – in a search, before any code, rather than in a retraction after.

Availability. Source, the pre-registration and literature-gate record for §6, and every claim’s entry in the project ledger: <https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids> (lean_src/Villani.lean, docs/designs/ZERO_SOUND_LANDAU_DAMPING_PROPOSAL.md).

References

- [1] OpenAI. Finite time blowup for Navier–Stokes and Euler equations: Lean 4 formalization. <https://github.com/openai/NavierStokesAndEuler>, 2026. commit 8937a8f.
- [2] Anthropic. Fermat’s last theorem in Lean 4. <https://github.com/anthropics/fermats-last-theorem>, 2026. Research artifact, Apache-2.0; commit aa2d8b3. Vendors files derived from the Imperial College FLT project, listed in its `ATTRIBUTION.md`.
- [3] Jacob Bedrossian. Formalization of Landau damping in the Vlasov–Poisson equations in Lean, 2026. arXiv:2609.16801.

- [4] Xavier Callens. Known answers first: a certified Landau root, a Vlasov–Poisson solver with its failure modes pinned, zero sound in Lean, and what persistent homology does and does not add, 2026. Zenodo, doi:10.5281/zenodo.22881069.
- [5] Cédric Villani. Fisher information in kinetic theory, 2025. Lecture notes, Mathemata Summer School, Chania, July 2024; arXiv:2501.00925.
- [6] Cédric Villani. On the kinetic Fokker–Planck equation in curved geometry, 2026. With F. Debbasch and Y. Ollivier; arXiv:2608.14904.
- [7] Cyril Imbert, Luis Silvestre, and Cédric Villani. On the monotonicity of the Fisher information for the Boltzmann equation. *Inventiones mathematicae*, 243:127–179, 2025. arXiv:2409.01183.
- [8] Clément Mouhot and Cédric Villani. On Landau damping. *Acta Mathematica*, 207:29–201, 2011.
- [9] Cédric Villani. Hypocoercivity. *Memoirs of the American Mathematical Society*, 202(950), 2009. arXiv:math/0609050.
- [10] Laurent Desvillettes, Clément Mouhot, and Cédric Villani. Celebrating Cercignani’s conjecture for the Boltzmann equation. *Kinetic and Related Models*, 4:277–294, 2011.
- [11] John Lott and Cédric Villani. Ricci curvature for metric-measure spaces via optimal transport. *Annals of Mathematics*, 169:903–991, 2009.
- [12] John Lott and Cédric Villani. Weak curvature conditions and functional inequalities. *Journal of Functional Analysis*, 245:311–333, 2007.
- [13] Yann Ollivier and Cédric Villani. A curved Brunn–Minkowski inequality on the discrete hypercube, or: what is the Ricci curvature of the discrete hypercube? *SIAM Journal on Discrete Mathematics*, 26(3):983–996, 2012. arXiv:1011.4779.
- [14] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, second edition, 2006. Section 4.4, “The second law of thermodynamics”.
- [15] Lean FRO. Comparator: a trustworthy judge for Lean proofs. <https://github.com/leanprover/comparator>, 2026.
- [16] H. Godfrin, M. Meschke, H.-J. Lauter, A. Sultan, H. M. Böhm, E. Krotscheck, and M. Panholzer. Observation of a roton collective mode in a two-dimensional Fermi liquid. *Nature*, 483:576–579, 2012.
- [17] I. Ya. Pomeranchuk. On the stability of a Fermi liquid. *Soviet Physics JETP*, 8:361–362, 1958. Zh. Eksp. Teor. Fiz. 35, 524 (1958). Cited via secondary sources (Baym & Pethick; Wikipedia’s “Pomeranchuk instability”); the primary Russian text was not directly consulted.
- [18] Evgeni E. Kolomeitsev and Dmitry N. Voskresensky. Scalar quanta in Fermi liquids: zero sounds, instabilities, Bose condensation, and a metastable state in dilute nuclear matter. *European Physical Journal A*, 52:362, 2016. arXiv:1610.09748.
- [19] Xavier Callens. Formalizing quantum-fluid structure in Lean 4, 2026. Zenodo, doi:10.5281/zenodo.22855581.